

STAT 1 Intro to course topics

Main subject	#	Topics	Description
Basics	1	Basic definitions and theory of probability	Basic rules for calculating probabilities
Statistical modeling	2	Discrete random variables and probability distributions	Complete and summarizing description of discrete probability distributions for a single variable (countable number of outcomes). Examples of common discrete distributions
	3	Continuous random variables and probability distributions	Complete and summarizing description of continuous probability distributions for a single variable (non-countable infinitely many outcomes). Examples of common continuous distributions
	4	Joint probability distributions	Probability distributions for several (dependent) variables
Estimation of model parameters	5	Descriptive statistics Sampling distributions Point estimation	Visualization of data ("the sample"). Quantitative description of data with a few characteristic numbers. Point-estimation : "statistically qualified guess" on <u>the value</u> of unknown parameters for an assumed model for the sample. Properties for estimators
	6	Interval estimation (confidence intervals)	Interval-estimation : A statistically qualified <u>interval</u> containing the unknown parameter with a specified probability (frequency). Two-sided and one-sided intervals
Statistical hypothesis testing	7	Single-sample tests	Inference : Statistical <u>testing</u> if an unknown parameter should supposedly be equal to or greater/smaller than a specified value.
	8	Two-sample tests	Inference : Statistical <u>testing</u> for comparison of unknown parameters for two different samples.
Correlation, regression and prediction	9	Regression models: simple linear, multiple linear, polynomial and non-linear	Analysis of possible functional (linear or non-linear) connection for pair-wise (x,y)-observations. Estimation of model parameters
	10	Inference for simple linear regression	Inference for model parameters. Prediction of new observations
Comparison of several samples	11	ANOVA 1: One-factor analysis of variances	Comparison of several samples grouped by <u>a single categorical variable</u>
	12	ANOVA 2: Two-factor analysis of variances	Comparison of several samples grouped by <u>two categorical variables</u>